

§2.2

Histograms

The picture that shows you the shape of your data

Today's Goal

- Create frequency histograms and relative frequency histograms
- Use Excel to create histograms from raw data
- Identify common distribution shapes (symmetric, skewed, uniform)
- Interpret histograms to describe data distributions

What You Will Be Graded On

LT.5 Graphical Summaries of Data

I understand the different types of graphical displays for data (e.g., bar graphs, histograms, boxplots, etc.). I can interpret and create appropriate graphical summaries for various types of data.

You Already Have the Table

Forty students said how many minutes their bus ride to campus takes. Here is the grouped frequency table for those forty times.

Time (minutes)	10–24	25–39	40–54	55–69	70–84
Frequency	7	15	10	5	3

Lopsided, or even?

You can work it out from the numbers. A picture just tells you.

A Table Drawn as Bars

Definition 2.2.1

A **histogram** is a visual representation of a dataset consisting of bars of equal width. The horizontal axis is labeled with what your data represents. The vertical axis is labeled with the frequency (or relative frequency). The bars are adjacent to one another (unless there is a gap in the data). The data is divided into **bins** (or intervals) which are used to place the bars along the horizontal axis.

Numbers Only

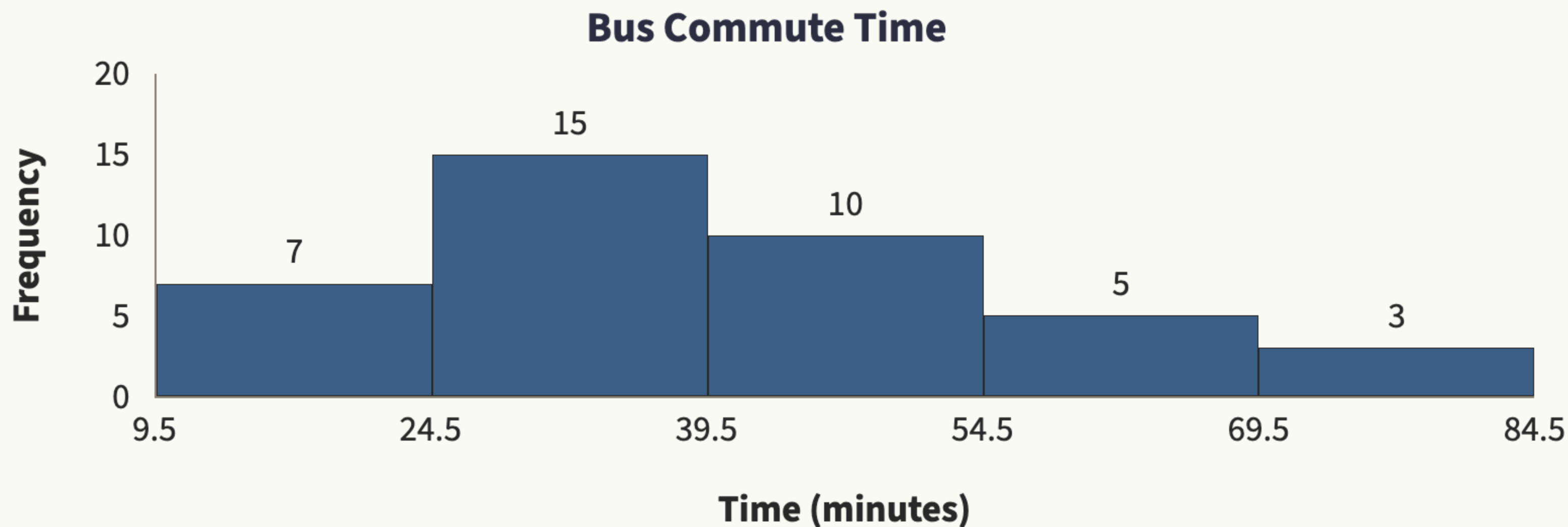
ASIDE

Histograms are *only* used for quantitative data.

The horizontal axis of a histogram is a number line, and each bin is an interval on it. Eye color has a frequency table, but blue and brown are not numbers, so there is no axis to place them on and no interval to make a bin from.

From Table to Bars

Example 1. Draw the histogram for the grouped bus-commute table.



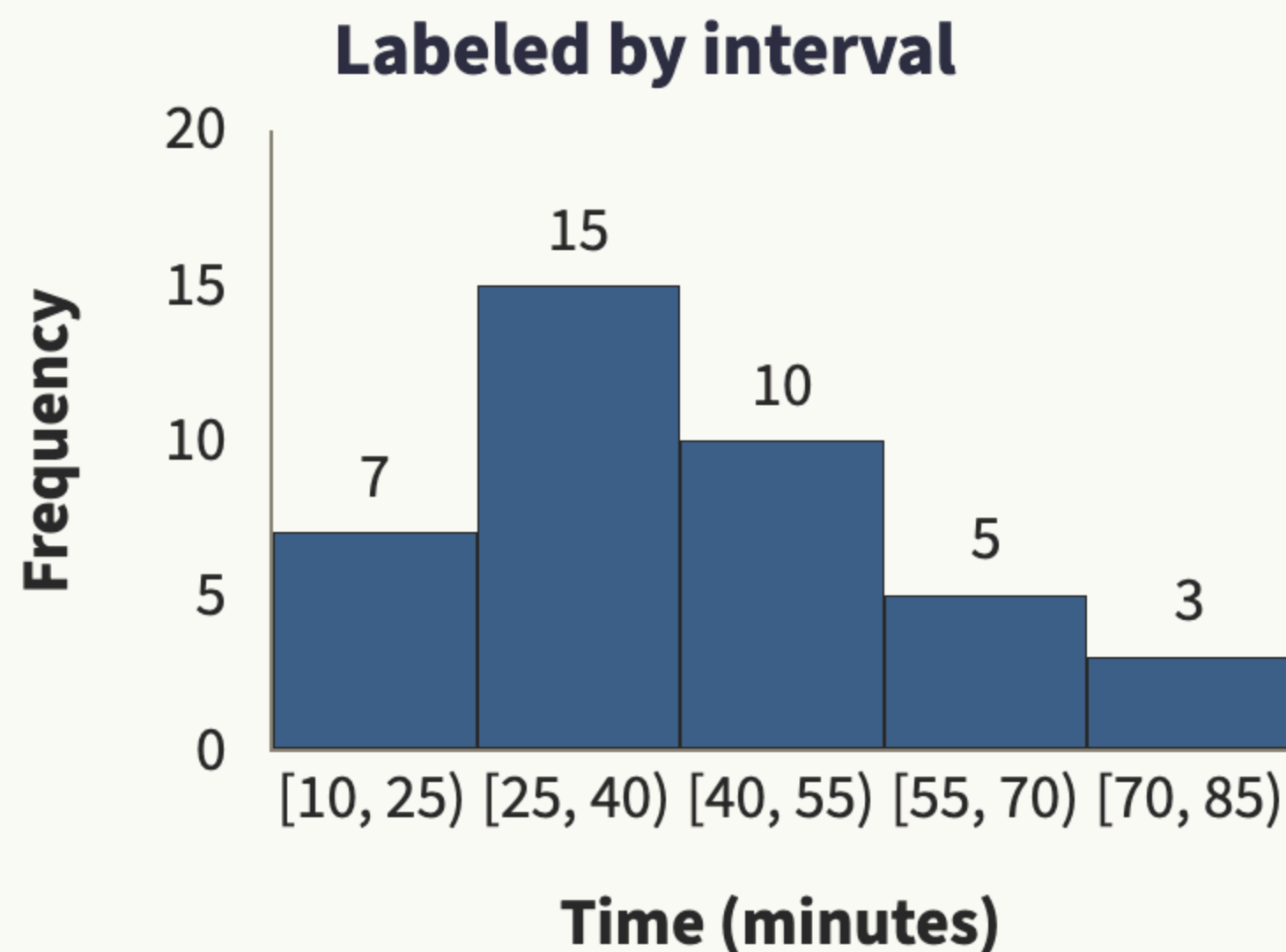
The bars touch because no class is empty.

Two Decisions, and You Are Done

1. **Where each bar starts and stops.** Mark the *class boundaries*, not the class limits: 9.5, 24.5, 39.5, 54.5, 69.5, 84.5.
2. **How tall each bar is.** Read the frequency out of the table.

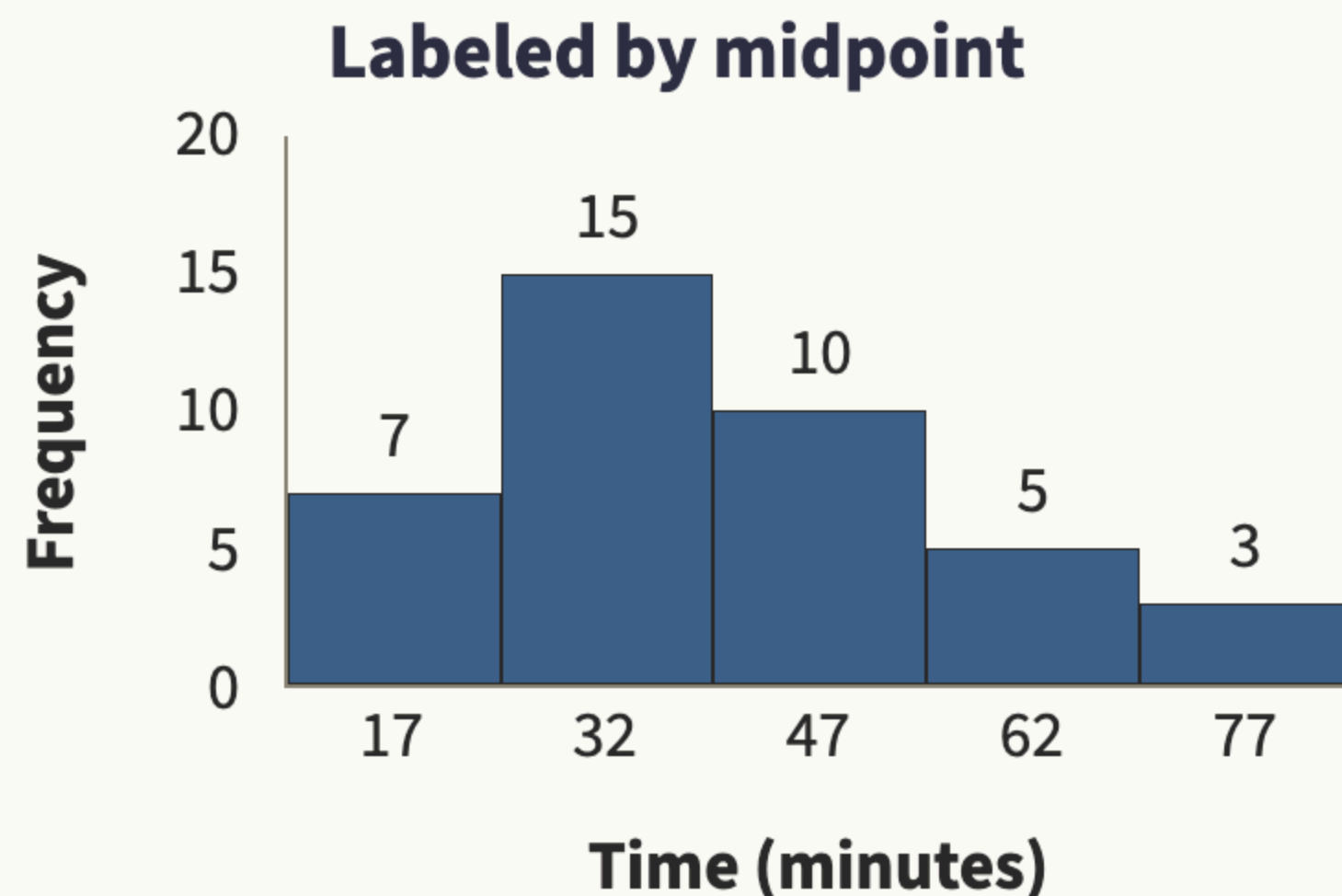
The bars touch because no class is empty. The boundaries decide where each bar starts and stops: the limits 24 and 25 leave a gap between them, and the boundary 24.5 is the single edge where the two bars meet.

Three Ways to Label the Axis



Each bar named by the interval it covers.

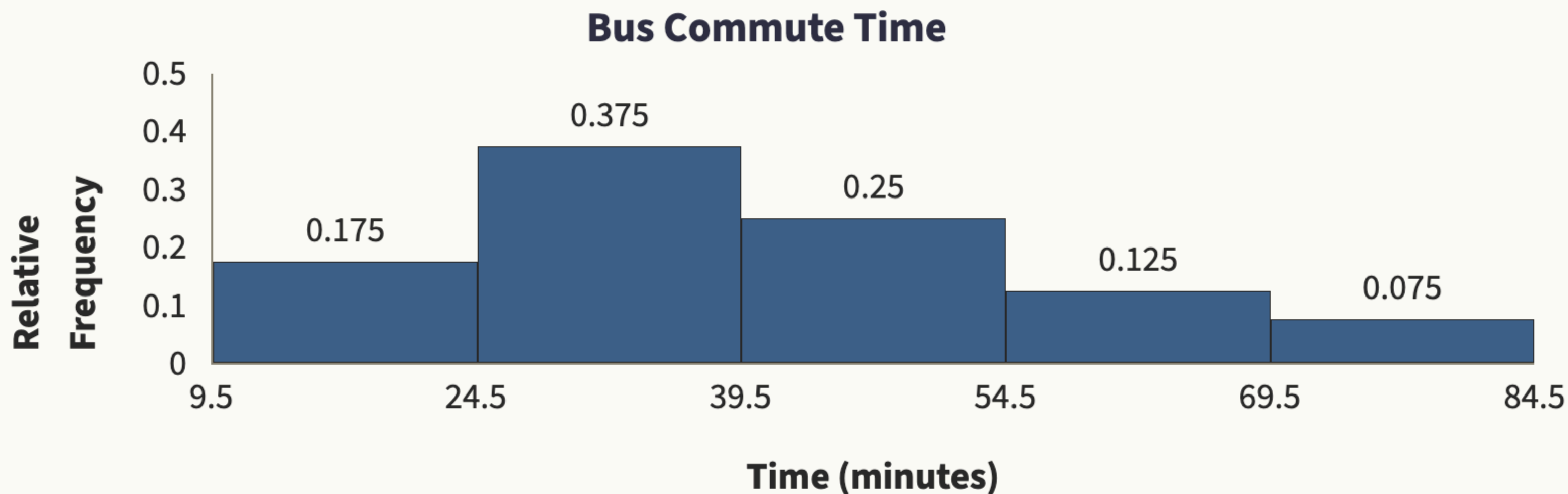
$[10, 25)$ means 10 up to but not including 25.



Each bar named by its midpoint. Rare, and harder to read.

Swap Counts for Proportions

Example 2. Draw the same data as a **relative frequency histogram**.



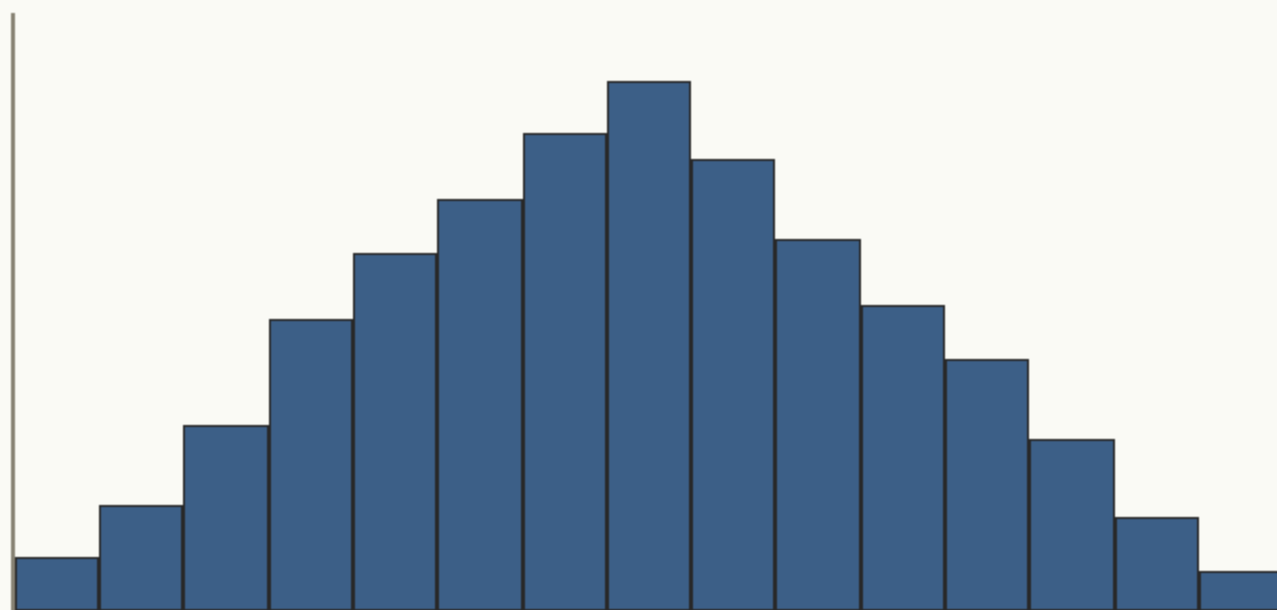
Same shape. Only the vertical axis changed, from counts to proportions.

Four Things You Can See

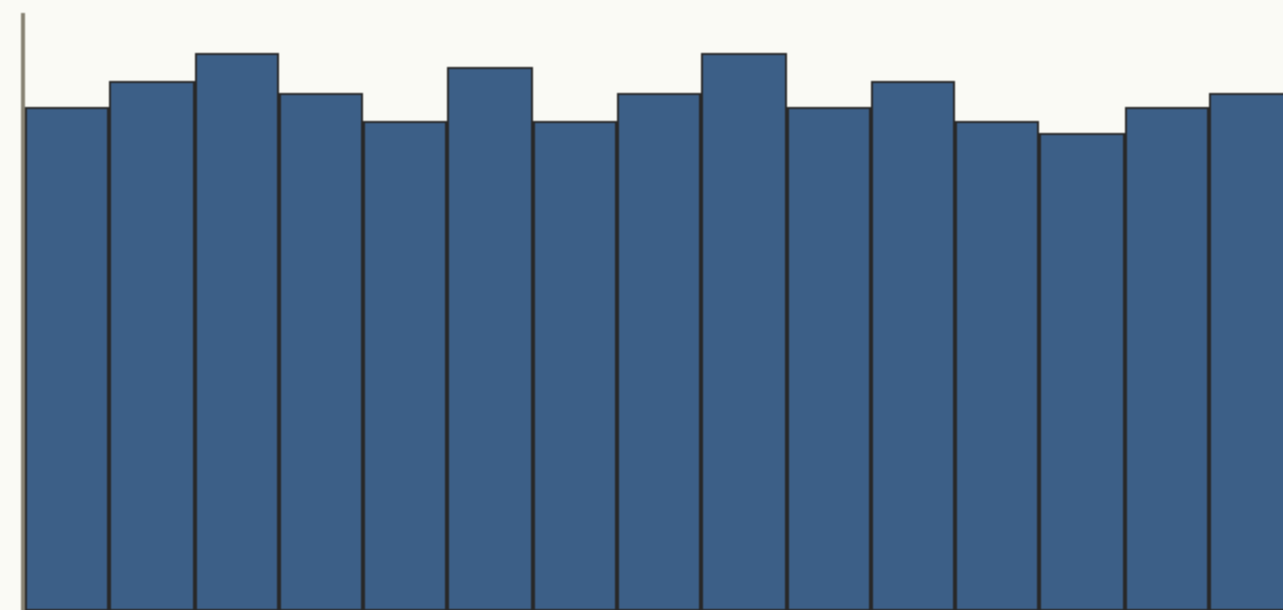
- the location of the *center* of the data
- the *spread* of the data
- any *outliers*
- the overall *shape* of the distribution of the data

Four shapes come up again and again. Learn to name them on sight.

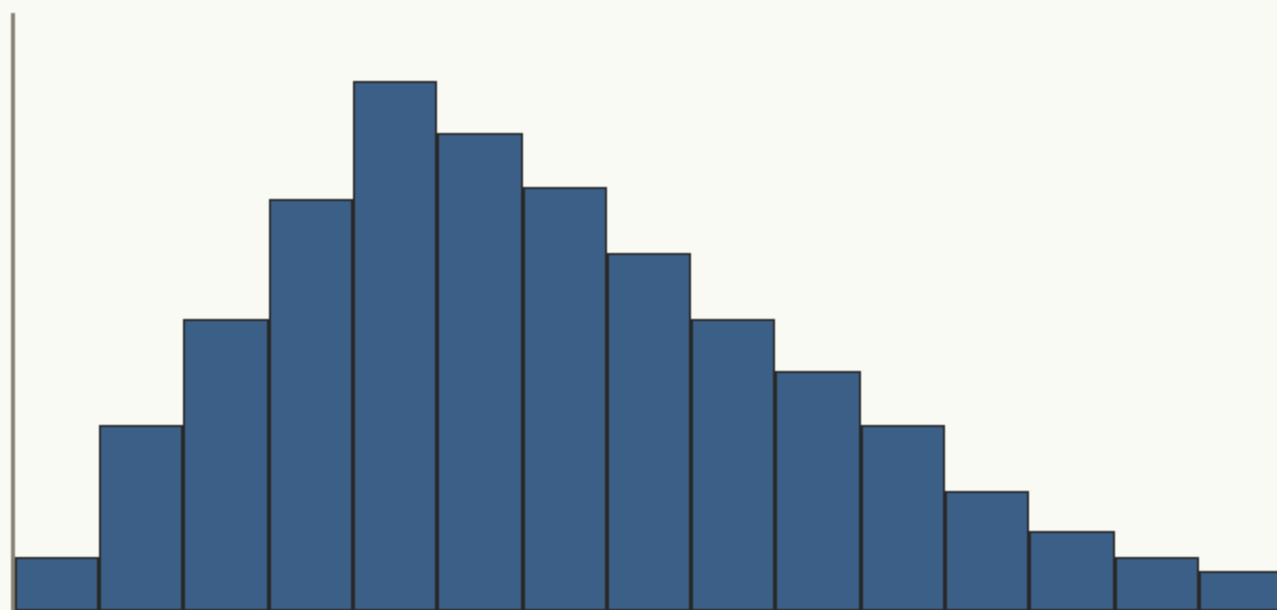
The Four Shapes



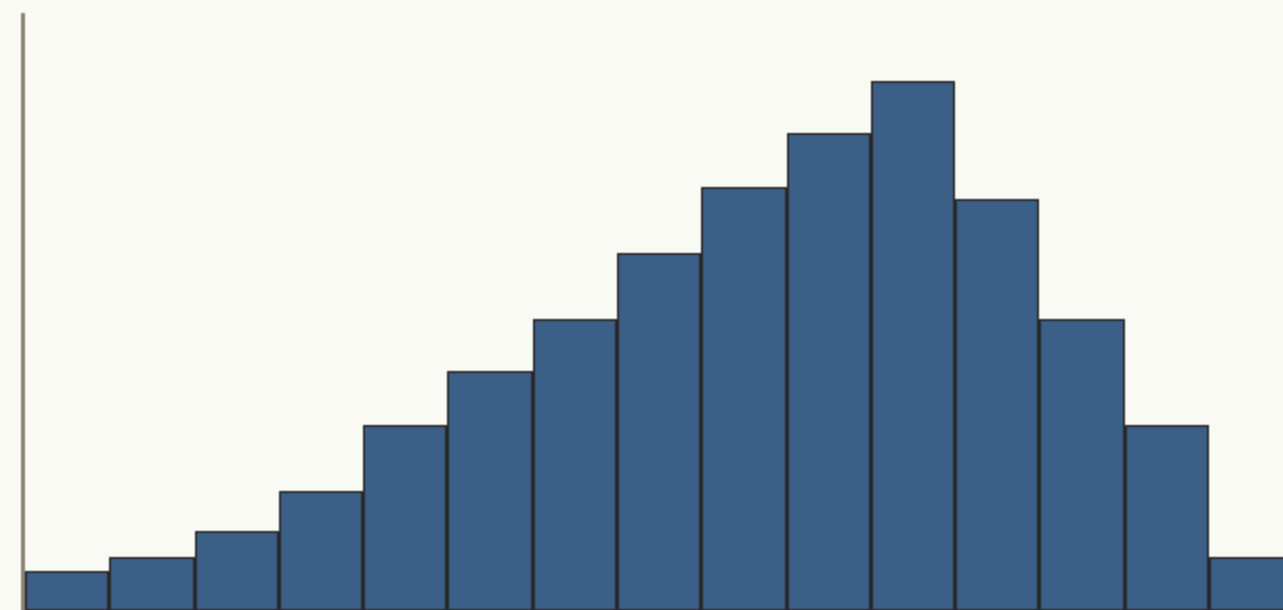
Bell-shaped



Uniform



Skewed to the right



Skewed to the left

One Peak, Even Sides

Definition 2.2.8

A distribution is **(approximately) bell-shaped** when its histogram is roughly symmetric and rises to a single peak near the center, then falls off symmetrically on both sides. This is also referred to as an **(approximately) normal distribution**; the formal definition of a normal distribution is given in a later chapter.

Flat Across the Top

Definition 2.2.10

A distribution is **(approximately) uniform** when the height of each bar in its histogram is approximately the same. Every value (or every interval) occurs with roughly the same frequency.

No peak, and no tail on either side.

Long Tail to the Right

Definition 2.2.12

A distribution is **skewed to the right** when its histogram is not symmetric and the right tail is longer than the left. The bulk of the data sits on the left, and a smaller number of larger values stretch the distribution out to the right.

Long Tail to the Left

Definition 2.2.14

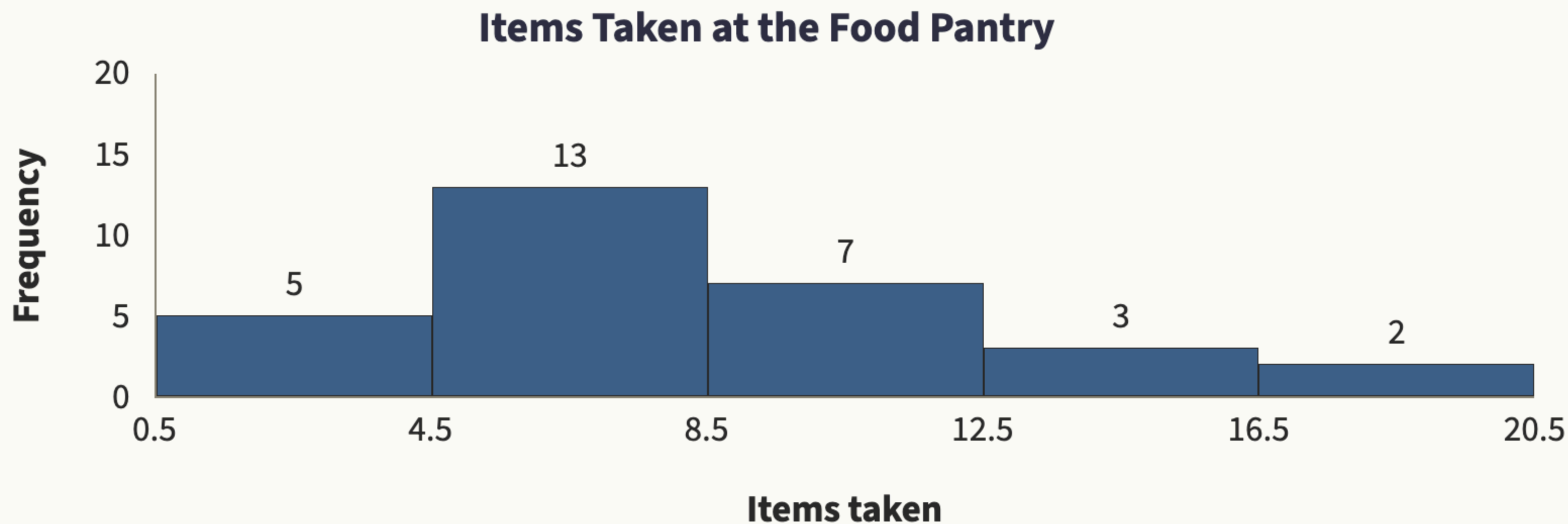
A distribution is **skewed to the left** when its histogram is not symmetric and the left tail is longer than the right. The bulk of the data sits on the right, and a smaller number of smaller values stretch the distribution out to the left.

Now You Try: Draw It and Name It

You Try 2.2.1. Thirty food pantry visitors said how many items they took. Sketch the histogram, then name the shape.

Items taken	1–4	5–8	9–12	13–16	17–20
Frequency	5	13	7	3	2

You Try 2.2.1: Checking Your Answers



Rises to a peak, then a long tail to the right.

Boundaries 0.5, 4.5, 8.5, 12.5, 16.5, 20.5. The shape is **skewed to the right**.

Let Excel Draw It

Procedure 2.2.16 Creating a Histogram in Excel

1. Enter your data in a single column of an Excel spreadsheet.
2. Select all of your data.
3. Go to the **Insert** tab on the ribbon.
4. In the Charts group, click on **Insert Statistical Chart** (the icon looks like a histogram).
5. Select **Histogram** from the dropdown menu.
6. Excel will automatically create a histogram with default bin widths.

Setting the Bin Width

Procedure 2.2.16, continued

To customize your histogram:

- **Change bin width:**
 - *PC:* Right-click on one of the bin labels under a bar, select “Format Axis”, and under “Axis Options” set the bin width or number of bins.
 - *Mac:* Right-click on one of the bars, select “Format Data Series”, and set the bin width or number of bins.

Labels and Appearance

Procedure 2.2.16, continued

- **Add labels:** Right-click on the chart and select “Add Data Labels” to show frequencies on each bar.
- **Modify appearance:** Use the Chart Design and Format tabs to change colors, add titles, and adjust the overall look.

Excel’s automatic bins are usually wrong for your data. Set the width yourself before you read a shape off the picture.

Now You Try: In Excel

You Try 2.2.2. Build a histogram of these twenty exam scores in Excel with a bin width of 10. Which bin holds the most scores, and what is the shape?

62	71	75	68	83
79	88	72	91	77
65	80	74	86	70
78	95	73	81	69